

Tutorial 7: R Programming and Exploration on Statistical Distributions (STUDENT'S VERSION)

Theme: From R Commands to Random Variables, Distribution Functions and Simulation

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Programme: Nurturing Future Risk Managers and Actuaries: An AI-powered Journey for Gifted Secondary Students

Synergy & Comparison

The Logic of this Tutorial: Students will learn how R names distributions, how `d`, `p`, `q` and `r` functions work, how random samples behave, how empirical distributions approximate theoretical distributions, and how simulation explains major ideas such as the Poisson approximation and the Central Limit Theorem.

Key Definition / Result

Learning Objectives. By the end of this tutorial, students should be able to:

1. use R as a calculator, a graphing tool and a simulation tool;
2. explain the meaning of `d`, `p` and `r` distribution functions;
3. generate random samples from Uniform, Binomial, Poisson and Normal distributions;
4. compare sample statistics with theoretical population quantities;
5. visualise probability mass functions, density functions, cumulative distribution functions and empirical cumulative distribution functions;
6. explain why larger samples often produce more stable estimates;
7. use simulation to investigate tail risk and the Central Limit Theorem;
8. use AI tools responsibly to ask better coding and interpretation questions.

A. Quick R Warm-Up: Commands, Vectors and Comments

Synergy & Comparison

Why start with R basics again? Statistical distributions are not only formulas on paper. In actuarial work, we often need to generate data, store values, transform vectors, draw graphs and interpret outputs. Before exploring distributions, students should be comfortable with the small pieces of R code that make simulations possible.

Key Definition / Result

In R, we often use:

Idea	R notation	Example
Assignment	<code><-</code> or <code>=</code>	<code>sample_size <- 1000</code>
Combine values	<code>c()</code>	<code>c(1, 4, 2, 5, 8)</code>
Sequence	<code>:</code> or <code>seq()</code>	<code>1:5, seq(0, 1, by = 0.1)</code>
Repeat	<code>rep()</code>	<code>rep(1:3, each = 2)</code>
Extract values	<code>[]</code>	<code>x[1:3]</code>
Comment	<code>#</code>	<code># This line explains the code</code>

Comments are ignored by R but very important for human readers. Good comments explain the purpose of a line of code.

Suggested AI Prompt: Reviewing Basic R Commands

★ Core

“Act as a beginner-friendly R programming tutor. Please explain assignment, vectors, sequences, comments and vector extraction in R. Use simple examples related to risk or probability, and explain each line of code.”

Worked Example: A Simple Vector of Claim Amounts

★ Core

R Code:

```
# A small set of possible insurance claim amounts
claim_amounts <- c(1200, 500, 0, 2300, 800, 0, 1500)

# Show the first three claim amounts
claim_amounts[1:3]

# Count how many claims are positive
```

```

positive_claims <- claim_amounts > 0
sum(positive_claims)

# Calculate total claim amount
total_claim_amount <- sum(claim_amounts)

# Calculate average claim amount
average_claim_amount <- mean(claim_amounts)

total_claim_amount
average_claim_amount

```

Reflection Question: Readable Code

Why is it useful to use names such as `claim_amounts` and `average_claim_amount` instead of very short names such as `x` and `y`?

D. The R Distribution Family: d, p, q and r

Key Definition / Result

For many common distributions, R uses a consistent naming system:

`d` + distribution name, `p` + distribution name, `r` + distribution name.

Prefix	Meaning	Example
d	probability mass / density value	<code>dbinom(5, 6, 0.4)</code>
p	cumulative probability	<code>ppois(3, 2)</code>
r	random generation	<code>rnorm(5, 0, 1)</code>

For example, `norm` means Normal distribution, `unif` means Uniform distribution, `binom` means Binomial distribution, and `pois` means Poisson distribution.

Suggested AI Prompt: Understanding d, p, q and r

★★ Intermediate

“Act as an actuarial statistics tutor. Explain the difference between d, p, q and r functions in R. Use Uniform, Binomial, Poisson and Normal examples. For each example, explain the probability meaning before showing the R code.”

Worked Example: Four Distribution Commands**★★ Intermediate****R Code:**

```
# 1. Binomial probability:
# Probability that X = 5 when X follows Binomial(6, 0.4)
dbinom(5, size = 6, prob = 0.4)

# 2. Poisson cumulative probability:
# Probability that X <= 3 when X follows Poisson(2)
ppois(3, lambda = 2)

# Generate 5 random numbers from standard normal distribution
rnorm(5, mean = 0, sd = 1)
```

Common Mistake

For continuous distributions, `dnorm(x)` is not the probability that $X = x$. It is the height of the density curve at x . For a continuous random variable, the probability at one exact point is 0.

E. Exploring Four Important Distributions**Key Definition / Result**

We focus on four distributions that appear frequently in risk and actuarial modelling:

Distribution	R name	Main arguments	Typical use
Uniform	<code>unif</code>	<code>min, max</code>	simple random uncertainty
Binomial	<code>binom</code>	<code>size, prob</code>	number of successes/defaults/claims
Poisson	<code>pois</code>	<code>lambda</code>	count of rare events
Normal	<code>norm</code>	<code>mean, sd</code>	returns, approximations

Investigation / Activity: Generating and Comparing Samples**★★****Intermediate**

Task. For each of Uniform, Binomial, Poisson and Normal distributions, generate 1000 observations. Then calculate the sample mean and sample variance.

R Code:

```
set.seed(2026)
sample_size <- 1000

uniform_sample <- runif(sample_size, min = 0, max = 1)
binomial_sample <- rbinom(sample_size, size = 10, prob = 0.3)
poisson_sample <- rpois(sample_size, lambda = 3)
normal_sample <- rnorm(sample_size, mean = 0, sd = 1)

summarise_sample <- function(sample_values) {
  c(
    Mean = mean(sample_values),
    Variance = var(sample_values)
  )
}

summary_table <- rbind(
  Uniform = summarise_sample(uniform_sample),
  Binomial = summarise_sample(binomial_sample),
  Poisson = summarise_sample(poisson_sample),
  Normal = summarise_sample(normal_sample)
)

round(summary_table, 4)
```

Investigation / Activity: Visualising the Four Samples

★★ Intermediate

Task. Draw histograms for the four samples and compare their shapes.

R Code:

```
par(mfrow = c(2, 2))

hist(
  uniform_sample,
  main = "Uniform(0, 1)",
  xlab = "Value",
  col = "lightblue",
  border = "white"
)

hist(
  binomial_sample,
  main = "Binomial(10, 0.3)",
  xlab = "Number of Successes",
  col = "lightgreen",
  border = "white"
```

```

)

hist(
  poisson_sample,
  main = "Poisson(3)",
  xlab = "Number of Events",
  col = "lightyellow",
  border = "white"
)

hist(
  normal_sample,
  main = "Normal(0, 1)",
  xlab = "Value",
  col = "pink",
  border = "white"
)

par(mfrow = c(1, 1))

```

Reflection Question: Matching Distribution to Story

Which distribution would you use for the number of students absent from a class of 30 if each student independently has the same absence probability? Which distribution would you use for the number of accidents arriving at an insurance company in one day?

F. Empirical CDF and Theoretical CDF

Key Definition / Result

The **cumulative distribution function** of a random variable X is

$$F(x) = P(X \leq x).$$

The **empirical cumulative distribution function** estimates this from observed data.

In R, we can use:

```
plot(ecdf(sample_values))
```

to draw the empirical CDF.

Suggested AI Prompt: Empirical Versus Theoretical CDF ** Intermediate

“Please explain the difference between an empirical CDF and a theoretical CDF. Then write beginner-friendly R code to generate a normal sample, draw its empirical CDF, and overlay the theoretical normal CDF.”

Worked Example: Empirical CDF of a Normal Sample ** Intermediate**R Code:**

```
set.seed(2026)
normal_sample <- rnorm(100, mean = 0, sd = 1)

plot(
  ecdf(normal_sample),
  main = "Empirical CDF and Theoretical Normal CDF",
  xlab = "x",
  ylab = "Cumulative Probability",
  col = "darkblue"
)

x_grid <- seq(from = -3, to = 3, by = 0.02)
theoretical_cdf <- pnorm(x_grid, mean = 0, sd = 1)

lines(
  x_grid,
  theoretical_cdf,
  col = "red",
  lwd = 2
)

legend(
  "bottomright",
  legend = c("Empirical CDF", "Theoretical Normal CDF"),
  col = c("darkblue", "red"),
  lwd = c(1, 2),
  bty = "n"
)
```

Common Mistake

A larger sample does not guarantee that every sample statistic is exactly equal to the theoretical value. It only makes large deviations less likely.

G. Binomial Distribution and Poisson Approximation

Key Definition / Result

If

$$X \sim \text{Binomial}(n, p),$$

then X counts the number of successes in n independent trials, each with success probability p .

When n is large, p is small, and $np = \lambda$ stays moderate, the Binomial distribution can be approximated by a Poisson distribution:

$$\text{Binomial}(n, p) \approx \text{Poisson}(\lambda), \quad \lambda = np.$$

Worked Example: Default Counts as a Binomial Model ** Intermediate

Question. Suppose 250 clients each have a 6% chance of default. What is the probability that exactly 30 clients default? What is the probability that more than 30 clients default?

R Code:

```
number_of_clients <- 250
default_probability <- 0.06

prob_exactly_30 <- dbinom(
  x = 30,
  size = number_of_clients,
  prob = default_probability
)

prob_more_than_30 <- 1 - pbinom(
  q = 30,
  size = number_of_clients,
  prob = default_probability
)

prob_exactly_30
prob_more_than_30
```

Investigation / Activity: Seeing Binomial Become Poisson *** Advanced

Task. Fix $\lambda = 1$. Let $p = 1/n$, so that $np = 1$. Plot Binomial probabilities for increasing n , and compare them with Poisson probabilities.

R Code:

```

lambda <- 1
x_values <- 0:8
n_values <- 2 * (1:8)
p_values <- lambda / n_values
plot(
  x_values,
  dpois(x_values, lambda),
  type = "h",
  lwd = 4,
  col = "black",
  main = "Binomial Approximation to Poisson",
  xlab = "Number of Events",
  ylab = "Probability"
)
for (i in 1:length(n_values)) {
  lines(
    x_values,
    dbinom(x_values, size = n_values[i], prob = p_values[i]),
    type = "b",
    col = i + 1,
    pch = 16
  )
}
legend(
  "topright",
  legend = c(
    "Poisson(lambda = 1)",
    paste0("Binomial(n = ", n_values, ", p = ", round(p_values, 3), ")")
  ),
  col = c("black", 2:(length(n_values) + 1)),
  lty = 1,
  pch = c(NA, rep(16, length(n_values))),
  cex = 0.7,
  bty = "n"
)

```

Reflection Question: Why Rare Events Matter

Why might the Poisson distribution be useful for modelling the number of rare events, such as serious accidents, insurance claims of a special type, or system failures?

H. Normal Distribution and Tail Risk

Key Definition / Result

If

$$X \sim N(\mu, \sigma),$$

then X has mean μ and standard deviation σ . In R:

Command	Meaning	Example
<code>dnorm(x, mean, sd)</code>	density height	<code>dnorm(0,0,1)</code>
<code>pnorm(q, mean, sd)</code>	$P(X \leq q)$	<code>pnorm(5,3,2)</code>
<code>rnorm(n, mean, sd)</code>	random sample	<code>rnorm(100,3,2)</code>

Worked Example: Normal Probability

★★ Intermediate

Question. If $X \sim N(3, 2)$, find $P(X < 5)$ and $P(0 < X < 5)$.

R Code:

```
mean_X <- 3
sd_X <- 2
prob_less_than_5 <- pnorm(5, mean = mean_X, sd = sd_X)
prob_between_0_and_5 <- pnorm(5, mean = mean_X, sd = sd_X) -
  pnorm(0, mean = mean_X, sd = sd_X)
prob_less_than_5
prob_between_0_and_5
```

I. Simulation, Sample Size and Stability

Key Definition / Result

A simulation uses random numbers to imitate uncertain outcomes. When we repeat a simulation many times, we can estimate quantities such as:

1. expected loss;
2. probability of exceeding a threshold;
3. tail averages;
4. distribution shape.

The larger the simulation size, the more stable the estimates usually become.

J. Central Limit Theorem Through Simulation

Key Definition / Result

The **Central Limit Theorem** says that, under suitable conditions, the distribution of sample means becomes approximately Normal when the sample size is large, even if the original data are not Normal.

If $X_1, \dots, X_n$ are independent observations from the same distribution with mean μ and standard deviation σ , then

$$\bar{X} = \frac{X_1 + \dots + X_n}{n}$$

has mean μ , and its standard deviation is

$$\frac{\sigma}{\sqrt{n}}.$$

This smaller standard deviation is called the **standard error**.

Suggested AI Prompt: Explaining the Central Limit Theorem

Advanced

“Explain the Central Limit Theorem to gifted secondary school students using simulation in R. Use Binomial, Poisson and Uniform examples. Emphasise that the distribution of the sample mean can look more Normal even when the original distribution is not Normal.”

Investigation / Activity: Distribution Versus Distribution of Sample Means

*** Advanced

Task. Generate 1000 sample means. Each sample mean is calculated from 50 observations generated from a Binomial distribution. Compare the histogram of individual observations with the histogram of sample means.

R Code:

```
set.seed(2026)
number_of_repetitions <- 1000
sample_size_each_time <- 50
sample_means <- rep(0, number_of_repetitions)

for (repeat_index in 1:number_of_repetitions) {
  one_sample <- rbinom(
    n = sample_size_each_time,
    size = 10,
    prob = 0.2
  )
  sample_means[repeat_index] <- mean(one_sample)
```

```

    )
    sample_means[repeat_index] <- mean(one_sample)
  }

individual_observations <- rbinom(
  n = number_of_repetitions * sample_size_each_time,
  size = 10,
  prob = 0.2
)

par(mfrow = c(1, 2))
hist(
  individual_observations,
  main = "Individual Binomial Observations",
  xlab = "Value",
  col = "lightblue",
  border = "white"
)
hist(
  sample_means,
  main = "Sample Means",
  xlab = "Sample Mean",
  col = "lightgreen",
  border = "white"
)
par(mfrow = c(1, 1))

```

Investigation / Activity: Trying Different Original Distributions

Advanced

Task. Repeat the sample-mean experiment using Uniform, Poisson and Normal distributions.

R Code:

```

simulate_sample_means <- function(distribution_name,
                                   number_of_repetitions = 1000,
                                   sample_size_each_time = 50) {
  sample_means <- rep(0, number_of_repetitions)
  for (repeat_index in 1:number_of_repetitions) {
    if (distribution_name == "Uniform") {
      one_sample <- runif(sample_size_each_time, min = 0, max = 1)
    } else if (distribution_name == "Poisson") {
      one_sample <- rpois(sample_size_each_time, lambda = 3)
    } else if (distribution_name == "Normal") {
      one_sample <- rnorm(sample_size_each_time, mean = 0, sd = 1)
    }
    sample_means[repeat_index] <- mean(one_sample)
  }
}

```

```

    } else {
      stop("Unknown distribution name.")
    }
    sample_means[repeat_index] <- mean(one_sample)
  }
  return(sample_means)
}

uniform_means <- simulate_sample_means("Uniform")
poisson_means <- simulate_sample_means("Poisson")
normal_means <- simulate_sample_means("Normal")

par(mfrow = c(1, 3))
hist(uniform_means, main = "Means from Uniform", xlab = "Sample Mean",
     col = "lightblue", border = "white")
hist(poisson_means, main = "Means from Poisson", xlab = "Sample Mean",
     col = "lightyellow", border = "white")
hist(normal_means, main = "Means from Normal", xlab = "Sample Mean",
     col = "pink", border = "white")
par(mfrow = c(1, 1))

```

Reflection Question: The Power of Averages

Why does the distribution of sample means tend to look more regular than the distribution of individual observations?

K. AI-Assisted Interpretation and Critical Checking

Synergy & Comparison

AI as a learning partner: AI tools can help students understand R code, debug mistakes and explain statistical ideas. However, AI output should always be checked against the mathematical meaning and the actual R output.

Suggested AI Prompt: Asking AI to Explain Simulation Output** Insight

“I generated histograms for Uniform, Binomial, Poisson and Normal samples in R. I also generated histograms of sample means. Please explain what the plots show in student-friendly language. Tell me how this relates to sample size, and the Central Limit Theorem.”

Investigation / Activity: Critiquing an AI Explanation**** Insight**

Task. Ask an AI tool to explain the Central Limit Theorem simulation. Then check whether its answer includes all four of the following ideas:

1. The original data distribution and the sample-mean distribution are different objects.
2. Sample means tend to vary less than individual observations.
3. Larger sample size reduces the standard error.
4. Approximate Normality of sample means does not mean the original data are Normal.

L. Consolidation Questions**Investigation / Activity: Review Set A: R Programming Basics***** Core**

1. What is the purpose of the assignment operator `<=`?
2. What does `c(1,4,2,5,8)` create?
3. Why are comments useful in R code?
4. What is the difference between `1:5` and `seq(1,5,by=0.5)`?
5. Write one line of R code to calculate the mean of `c(2,5,7,9)`.

Investigation / Activity: Review Set B: Distribution Functions********Intermediate**

1. What does the prefix `d` mean in `dbinom()` or `dnorm()`?
2. What does the prefix `p` mean in `pnorm()`?
3. What does the prefix `r` mean in `runif()`?
4. Write R code to generate 1000 observations from $N(0,1)$.

Investigation / Activity: Review Set C: Simulation**** Intermediate**

1. Why do we use `set.seed()` in simulations?

2. Why might simulated values not be exactly equal to theoretical expectations?
3. Repeat the simulation using 100000 observations. What changes?

Investigation / Activity: Review Set D: Central Limit Theorem

Advanced

1. What is the difference between a distribution of observations and a distribution of sample means?
2. Why does averaging reduce variability?
3. What is the standard error of the sample mean?
4. Does the Central Limit Theorem say that all data are Normal? Explain.
5. Give one actuarial or risk-management situation where the Central Limit Theorem may be useful.

Investigation / Activity: Challenge Questions

*** Challenge

1. If two simulations use the same seed, why do they produce the same random numbers?
2. How could a poor choice of distribution affect risk measurement?
3. In what situation might a Poisson model be inappropriate for count data?
4. How would you explain the difference between randomness and model uncertainty?

End of Tutorial 7